10. The duration of flight travel from Chennai to London through British Airlines is approximately 11 hours. The airplane begins its journey on Sunday at 23:30 hours. If the time at Chennai is four and half hours ahead to that of London's time, then find the time at London, when will the flight lands at London Airport.

2.6 Sequences

Consider the following pictures.

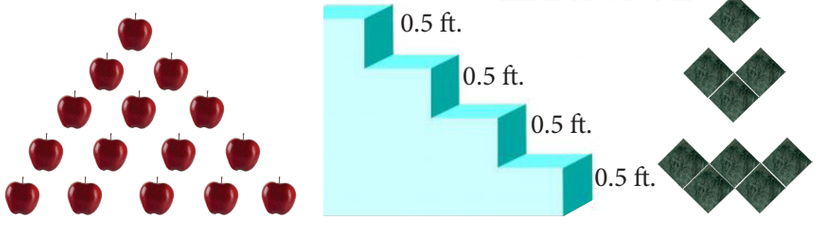


Fig.2.9

There is some pattern or arrangement in these pictures. In the first picture, the first row contains one apple, the second row contains two apples and in the third row there are three apples etc... The number of apples in each of the rows are 1, 2, 3, ...

In the second picture each step have 0.5 feet height. The total height of the steps from the base are 0.5 feet,1 feet, 1.5 feet,... In the third picture one square, 3 squares, 5 squares, ... These numbers belong to category called "Sequences".

Definition

A real valued sequence is a function defined on the set of natural numbers and taking real values.

Each element in the sequence is called a term of the sequence. The element in the first position is called the first term of the sequence. The element in the second position is called second term of the sequence and so on.

If the nth term is denoted by a_n, then a_1 is the first term, a_2 is the second term, and so on. A sequence can be written as $a_1, a_2, a_3, \ldots, a_n, \ldots$

Illustration

1. 1,3,5,7,... is a sequence with general term $a_n = 2n-1$. When we put $n = 1,2,3,\ldots,$ we get $a_1 = 1,\ a_2 = 3,\ a_3 = 5,\ a_4 = 7,\ldots$

2. $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$ is a sequence with general term $\frac{1}{n+1}$. When we put $n = 1,2,3,\ldots.$ we get $a_1 = \frac{1}{2},\ a_2 = \frac{1}{3},\ a_3 = \frac{1}{4},\ a_4 = \frac{1}{5},\ldots$

If the number of elements in a sequence is finite then it is called a Finite sequence. If the number of elements in a sequence is infinite then it is called an Infinite sequence.

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